Year 13 Further Maths - Elastic strings

Similar questions - Energy methods, friction and air resistance

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 One end of a light elastic string of natural length 0.5 m and modulus of elasticity 30 N is attached to a fixed point O which is 0.8 m vertically below a horizontal ceiling. The other end is attached to a particle P of mass 0.3 kg. P is held at rest at a point 1.2 m vertically below O and released. Assuming no obstruction as P passes O, find the speed of P when it first hits the ceiling. **[6]**

2 A light elastic string of natural length 1.4 m and modulus 70 N has one end attached to a fixed point O and the other to an object B of mass 2.5 kg. B is projected vertically upwards from O with speed $u\ \mathrm{m\,s^{-1}}$.

(a) State one modelling assumption needed. **[1]**

(b) The greatest height of B above O is 3.4 m. Find u. **[5]**

3 One end of a light elastic string of natural length 1.5 m and modulus 36 N is attached to a point O on a rough plane inclined at θ to the horizontal, $\sin\theta = 0.6$. The other end is attached to a particle P of mass 1.2 kg. The coefficient of friction is 0.5. P is placed at O and projected down a line of greatest slope with speed $3\ \mathrm{m\,s^{-1}}$. Find the distance P has travelled from O when it first comes to rest. You may assume that P does not reach the bottom of the plane. **[8]**

4 A bungee jumper of mass 70 kg steps off a high platform with an elastic rope attached to her ankles. The air resistance is modelled as a constant force of 25 N. The rope has natural length 12 m and modulus of elasticity 1470 N. By considering energy, find the total distance she falls before first coming to instantaneous rest. **[7]**

Answers

Question 1

Initial extension 0.7 m: EPE $= \frac{30(0.7)^2}{2(0.5)} = 14.7$ J. At the ceiling P is 0.8 m above O, extension 0.3 m: EPE $= 2.7$ J, and P is 2.0 m above its start. $14.7 = 2.7 + 0.3g(2.0) + \frac{1}{2}(0.3)v^2$, $v^2 = 40.8$, $v = 6.39\ \mathrm{m\,s^{-1}}$

Question 2

(a) The string is light / B is a particle / no air resistance

(b) Extension at greatest height $= 3.4 - 1.4 = 2.0$ m, EPE $= \frac{70(2.0)^2}{2(1.4)} = 100$ J. $\frac{1}{2}(2.5)u^2 = 2.5g(3.4) + 100 = 183.3$, $u^2 = 146.6$, $u = 12.1\ \mathrm{m\,s^{-1}}$

Question 3

The string is slack for the first 1.5 m. Let the extension at rest be e. Work-energy: $\frac{1}{2}(1.2)(9) + 1.2g(0.6)(1.5+e) - 0.5(1.2g\cos\theta)(1.5+e) = \dfrac{36e^2}{2(1.5)}$. Friction $= 0.5(1.2)(9.8)(0.8) = 4.704$ N, $mg\sin\theta = 7.056$ N. $5.4 + 2.352(1.5+e) = 12e^2$, $12e^2 - 2.352e - 8.928 = 0$, $e = 0.966$. Distance $= 1.5 + 0.966 = 2.47$ m

Question 4

Let her total fall be d (rope extension $d-12$). $70g\,d - 25d = \dfrac{1470(d-12)^2}{2(12)}$, i.e. $661d = 61.25(d-12)^2$. This simplifies to $61.25d^2 - 2131d + 8820 = 0$, whose relevant root is $d = 30.0$ m